

MINISTRY OF AGRICULTURE

MoA Procurement Tracking System

Software Design Document (SDD)

One-Month MVP - Version 1.2

Updated workflow, data design and UML diagrams

Item	Details
Document status	Draft for technical and stakeholder review
Confirmed actors	Procurement Officer, Procurement Director, Management Committee Member, Administrator
Selected stack	Next.js + TypeScript; Node.js + Express; Prisma ORM; PostgreSQL
Date	3 August 2026

Prepared for the MoA development team, Procurement Directorate and ICT technical review

Document Control

Item	Details
Document title	MoA Procurement Tracking System - Software Design Document
Version	1.2
Status	Draft
Owner	Ministry of Agriculture Procurement Directorate / ICT
Purpose	Define the implementation design for the updated one-month MVP.
Change basis	Stakeholder clarifications on project assignment, category-specific plans, two-level approval, generated references, and removal of document/audit-history features.

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1. Document Introduction

1.1 Purpose

This Software Design Document defines how the updated MoA Procurement Tracking System MVP will be implemented. It converts the confirmed business workflow into architecture, modules, domain models, database structures, APIs, user interfaces, security controls, reports, deployment and testing design.

1.2 Intended Readers

- Development team members implementing the Next.js frontend, Express API, Prisma data layer, reports and tests.
- MoA ICT staff reviewing security, infrastructure, deployment, backup and maintainability.
- Procurement Directorate representatives confirming that the system reproduces the essential Excel planning and reporting process.
- Testers preparing integration, role-access and user-acceptance evidence.

1.4 Scope

The MVP is an internal procurement planning, approval, progress-tracking, contract follow-up and reporting system. It does not perform electronic tender publication, bid submission, bid opening, bid evaluation, supplier self-service, online payment, digital signatures, or live integration with e-GP, IFMIS, banks or donor portals.

Supporting-document upload and user-facing audit history are explicitly outside the updated scope. The application still keeps normal technical logs for diagnosing errors, and it stores explicit business records such as Director decisions, committee votes and activity revisions because those records are required by the workflow.

1.5 Key Definitions

Term	Meaning
Project	A program or initiative created by the Procurement Director and assigned to one or more Procurement Officers.
Procurement Plan	A category-specific plan for an assigned project and fiscal year or planning period.
Procurement Activity	One Excel-equivalent procurement entry under a plan, such as procurement of vehicles or a consultancy assignment.
Activity Stage	A method-specific milestone such as advertisement, opening, evaluation, signing or completion.
Director Review	The Director decision to Approve, Return for Correction or Reject a submitted plan.
Committee Vote	One Approve or Reject decision recorded by an active member of the five-member Management Committee.
Final Approval	Automatic plan status reached when at least three committee members approve.
Replan	A change to a current target date that preserves the original date and requires a reason.
Canonical date	The single Gregorian business date stored in the database and converted for Ethiopian-calendar display.

2. Updated Business and Workflow Context

2.1 Core Hierarchy

The design uses the following hierarchy: Organization -> Project -> Procurement Plan -> Procurement Activity -> Activity Stage -> Contract and Payments, when applicable.

- **Project:** Created by the Procurement Director and assigned to Procurement Officers.
- **Plan:** Created by an assigned Officer after selecting the project. A plan has one category: Goods, Works, Non-Consulting Services or Consultancy Services.
- **Procurement Activity:** A main planning/reporting record under the plan. Its fields depend on plan category and procurement method.
- **Activity Stage:** A generated roadmap milestone with original, revised and actual dates, status, delay and remarks.

2.2 Roles and Responsibilities

Actor	Main actions	Restrictions
Procurement Officer	Open assigned projects; create category-specific plans; add activities and stages; update progress; replan; register contracts/payments; submit plans; generate scoped reports.	Cannot create projects, work in unassigned projects, approve own plans or vote unless separately assigned as a committee member.

Actor	Main actions	Restrictions
Procurement Director	Create/update projects; assign Officers; review submitted plans; approve/return/reject; monitor committee progress; view Ministry-wide operational data; receive replan alerts; generate reports.	Does not directly edit Officer-entered activity, stage, contract or payment data.
Management Committee Member	View Director-approved plans; submit one Approve or Reject vote; add comment; view voting progress.	Read-only for project, plan and activity data; one vote per plan.
Administrator	Manage users, roles, committee membership, organizations, controlled values, methods, stage templates and reference patterns.	Does not create or update procurement business records or approve plans.

2.3 Plan Lifecycle

Status	Allowed actions
DRAFT	Officer edits plan and activities.
SUBMITTED_TO_DIRECTOR	Plan locked for Officer; Director reviews.
RETURNED_FOR_CORRECTION	Officer corrects and resubmits.
REJECTED_BY_DIRECTOR	Workflow stops unless a new plan is created or an authorized reopening rule is added later.
DIRECTOR_APPROVED	Director decision stored; plan is sent to committee.
COMMITTEE_REVIEW	Five members vote; plan remains read-only.
FINALLY_APPROVED	Reached automatically when approval count is at least three.
REPLANNED	Approved operational target dates have been revised; original dates remain visible.
CLOSED / CANCELLED	End states controlled by permitted business rules.

2.4 Category-Specific Planning

A plan category is selected when the plan is created. The Activity form and exported report columns are then resolved from the combination of plan category and procurement method. This prevents one oversized form and ensures that the same information previously entered in the Excel workbooks is available in the system.

Plan category	Distinct information and milestone emphasis
Goods	Goods description, lot, amount, method, funding, review/qualification, advertisement, opening, evaluation, award, contract signing, optional letter of credit, delivery and handover.
Works	Works description, lot, qualification, review, invitation, opening, evaluation, award, signing, commencement, initial/revised/actual completion, physical progress and handover.
Non-Consulting Services	Service description, invitation/advertisement, receipt of bids or quotations, comparison/evaluation, award, signing, service start, completion and acceptance.
Consultancy Services	Assignment description, contract type, selection method, TOR, EOI, shortlist, RFP, technical/financial evaluation, negotiation, draft contract, award, signing and completion.

3. Design Goals and Principles

Principle	Design application
Familiar Excel-based workflow	Forms, tables and exports follow the logic and headings of the MoA planning, step-report, plan-vs-actual, contract and payment workbooks.
Simple role workflow	Only actions needed by the four confirmed actors are exposed.
Single entry and reuse	Project, plan and activity data is entered once and reused in roadmaps, approvals, contracts, dashboards and reports.
Category and method driven	Category determines fields and report layout; method determines the ordered activity-stage template.
Validation at every layer	Client validation improves usability; API services and database constraints remain authoritative.
Explicit approvals	Director reviews and committee votes are stored as business records with deterministic status rules.
No unnecessary features	No supporting documents, audit-history UI, e-tendering, online payment or external integration in the MVP.
Maintainable modular monolith	Clear Express modules share one PostgreSQL database through Prisma while remaining simple to deploy.

4. System Architecture

4.1 Architecture Style

The MVP uses a modular monolith with a separate Next.js frontend and Node.js/Express backend API. The backend is internally separated into routes, controllers, application services, domain services and Prisma repositories. All modules share one PostgreSQL database. This is simpler to deliver within one month than microservices while preserving clear module boundaries.

4.2 Selected Technology Stack

Layer	Selected technology	Use
Frontend	Next.js, TypeScript, React, App Router, MUI and MUI Data Grid Community	Role-aware pages, category-specific forms, roadmaps, tables, dashboards and report previews.
Backend API	Node.js, Express.js and TypeScript	REST endpoints, transactions, authorization, workflow and business rules.
Persistence	Prisma ORM	Typed PostgreSQL access, migrations and transactions.
Database	PostgreSQL	Projects, assignments, plans, activities, stages, approvals, votes, contracts, payments and alerts.
Authentication	Passport.js local strategy, database sessions, secure HTTP-only cookie, Argon2id	Sign-in, session and role-based access.
Validation	Shared Zod schemas plus domain services and database constraints	Consistent validation across UI and API.
Reports	PDFKit and ExcelJS	Server-generated PDF and XLSX exports.
Logging	Pino structured logging	Technical diagnostics, correlation IDs and safe production errors.
Deployment	Docker Compose behind Nginx on a MoA-controlled Linux server	Repeatable HTTPS deployment with persistent PostgreSQL storage.
Testing	Vitest, Supertest, React Testing Library and Playwright	Unit, API integration, component and end-to-end tests.

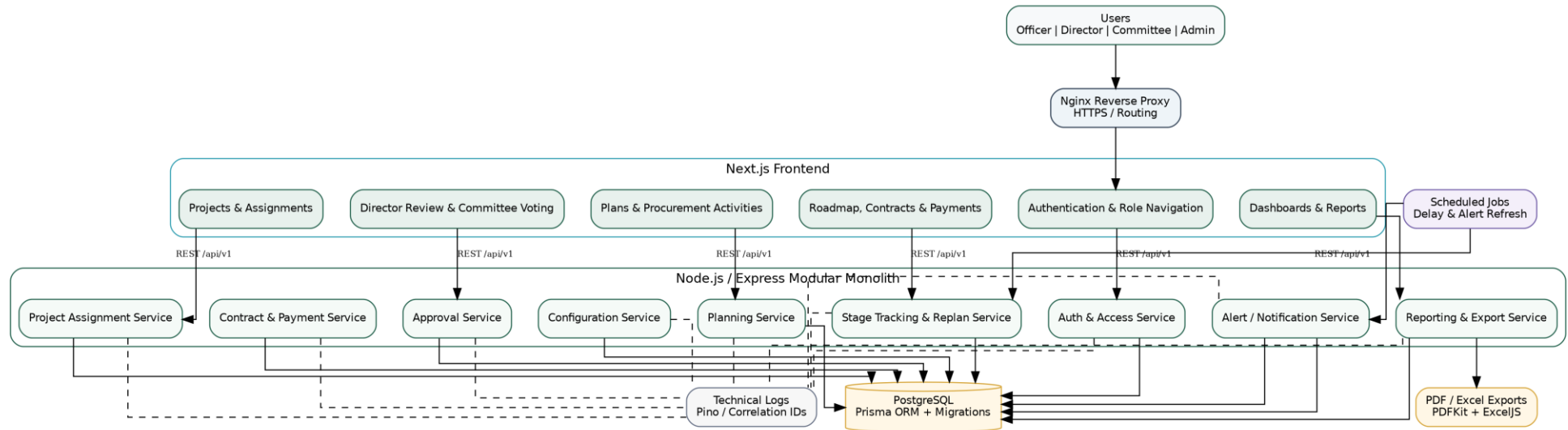


Figure 1. Updated component diagram - document storage and audit-history components are removed; project assignment and approval components are added.

4.4 Deployment Structure

Nginx terminates HTTPS and routes browser requests to the Next.js container and /api/v1 requests to the Express container. PostgreSQL uses persistent storage. Generated report files are streamed or stored only in a temporary report directory and are not treated as permanent procurement documents. Configuration and secrets are supplied through protected environment variables.

5. System Modules and Component Design

Module	Responsibility	Main services/components
Authentication and Access	Sign-in, password change, sessions, user-role and scope checks.	AuthController, AuthService, AuthorizationService, CurrentUserService.
Project and Assignment	Director project creation and Officer assignment.	ProjectController, ProjectService, ProjectAssignmentService.
Planning and Procurement Activities	Category-specific plans and activity records; generated references and method templates.	PlanService, ProcurementActivityService, ReferenceNumberService, ActivityTemplateService.
Approval Workflow	Officer submission, Director decision, committee membership, voting and final approval calculation.	PlanSubmissionService, DirectorReviewService, CommitteeVoteService, ApprovalStatusService.
Stage Tracking and Replanning	Original/revised/actual dates, statuses, progress, delay, revisions and Director alerts.	ActivityStageService, DelayCalculator, ProgressCalculator, ReplanService, AlertService.
Contract and Payment	Suppliers, contracts, amounts, dates, payment status and balances.	SupplierService, ContractService, PaymentService.
Dashboard and Reporting	Filtered summaries, report previews and PDF/XLSX export.	DashboardQueryService, ReportService, PdfExportService, ExcelExportService.
Administration	Users, roles, committee, organizations, methods, templates and controlled values.	UserAdminService, CommitteeAdminService, ConfigurationService.
Scheduled Processing	Daily delay recalculation and alert refresh.	Scheduler, DelayRefreshJob, AlertRefreshJob.

5.1 Dependency Rules

- Controllers call application services and do not directly update Prisma models.
- Planning may call ReferenceNumberService and ActivityTemplateService inside one transaction.
- Approval services are the only services allowed to change plan approval statuses.
- CommitteeVoteService enforces active membership, one vote per member and the three-approval threshold.
- Reporting uses read-only query services and does not modify procurement records.
- Technical logging is infrastructure-level; it is not a user-visible audit-history module.

6. UML and Process Diagrams

6.1 Use Case Diagram

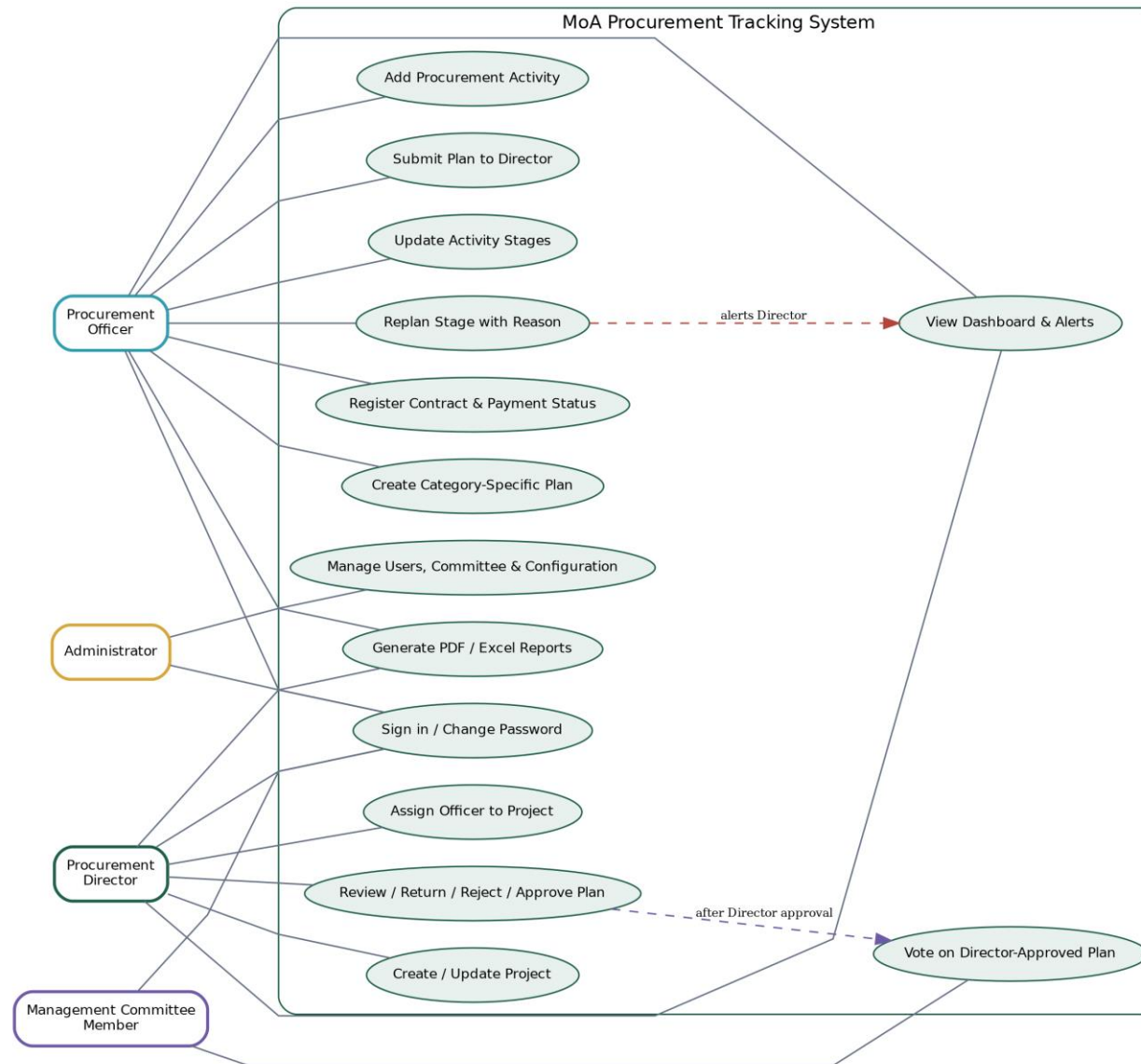


Figure 2. Use case diagram for the four confirmed actors.

6.2 Activity Diagram

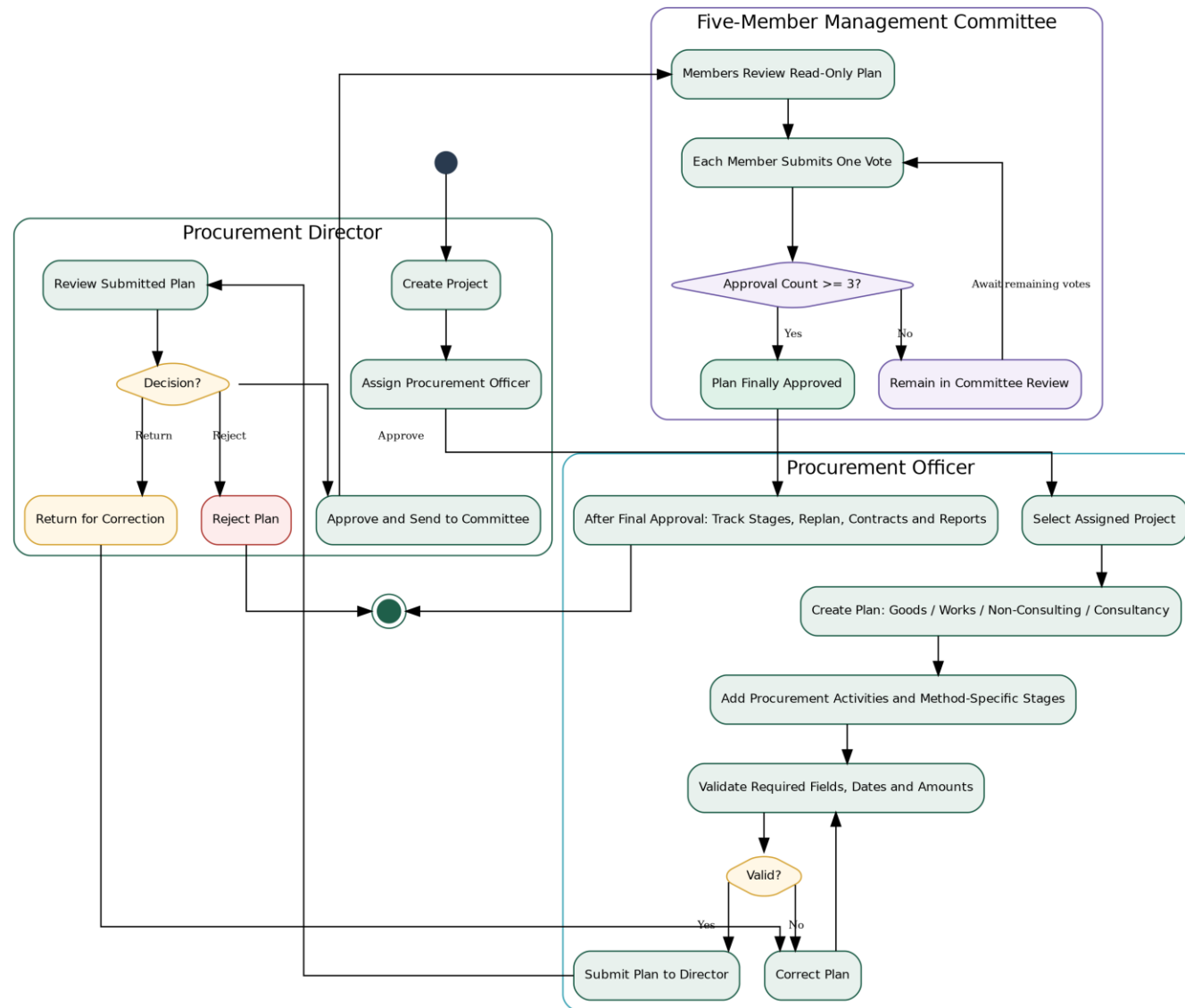


Figure 3. Plan creation and approval activity diagram.

6.3 Sequence Diagram

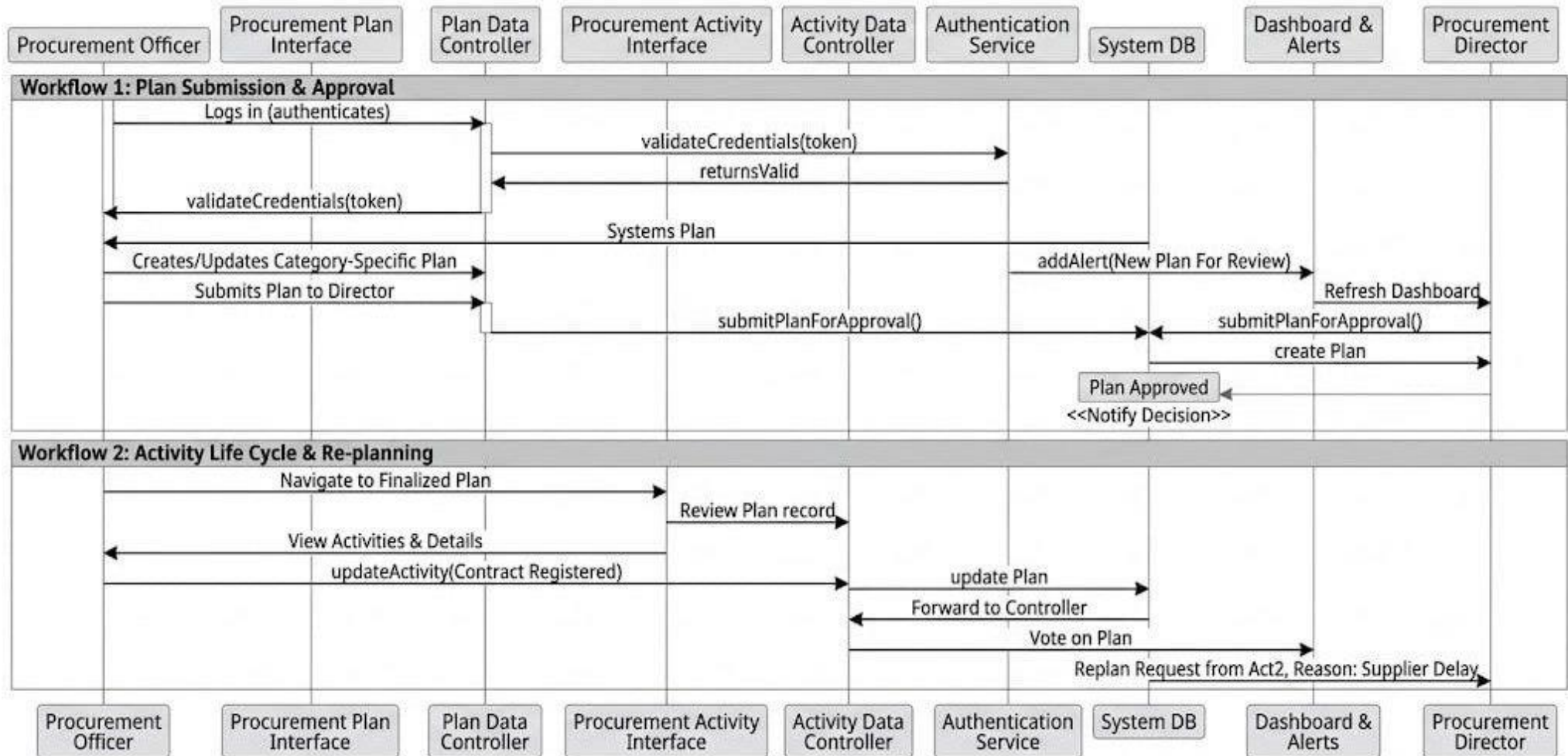


Figure 4. Plan submission, Director approval and committee voting sequence.

6.4 Class Diagram

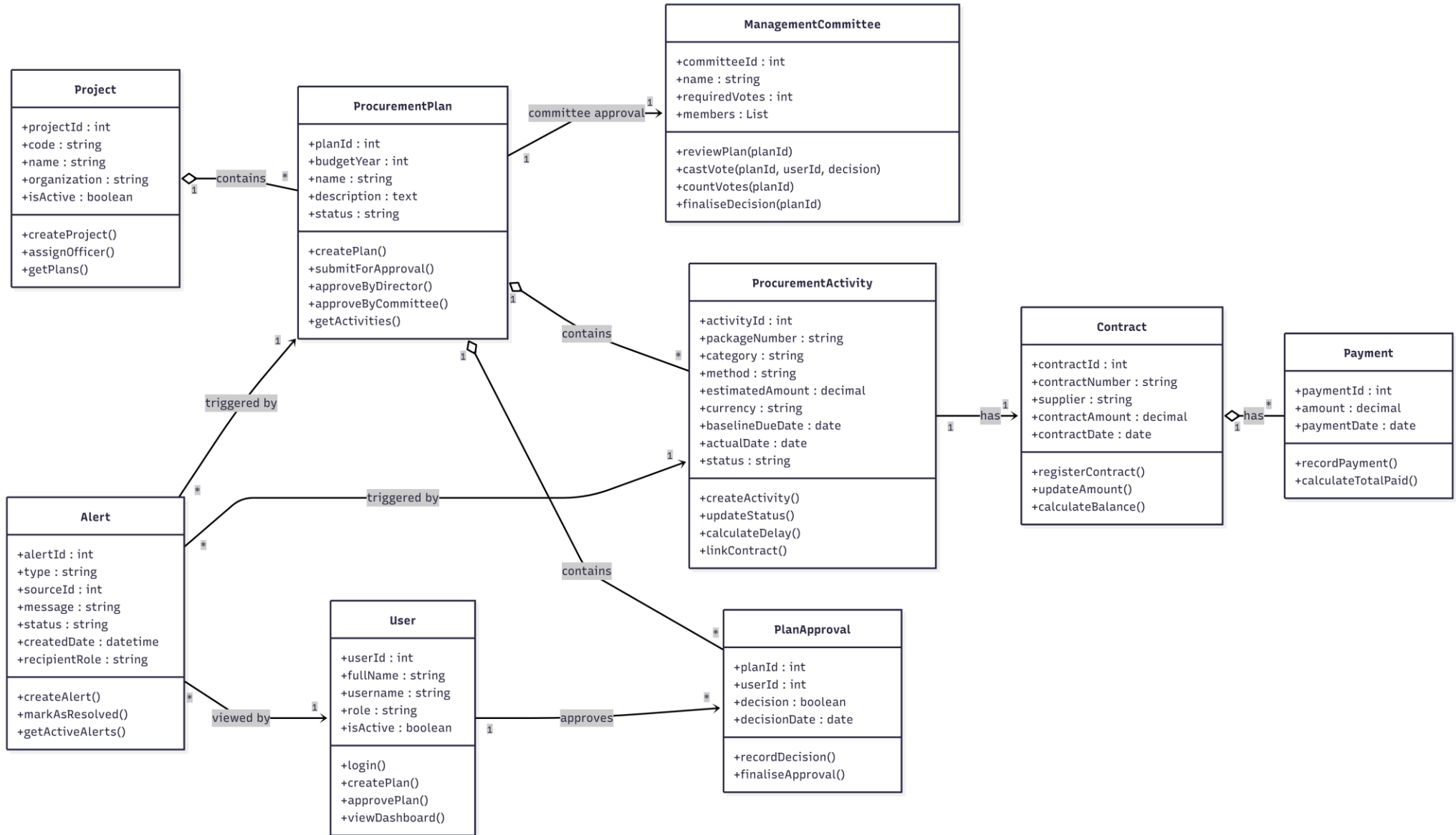


Figure 5. Updated domain class diagram.

7. Detailed Data Design

7.1 Database Conventions

- Business records use UUID primary keys; timestamps use timestampz in UTC; business dates use date.
- Money uses numeric(18,2); percentages use numeric(5,2).
- Foreign keys normally use ON DELETE RESTRICT. Active/inactive or status changes are preferred over deletion once records are operational.
- Generated references are normalized and protected by scoped unique indexes.
- Common procurement fields are relational. Category-specific metadata is stored in category_data JSONB and validated by category-specific Zod schemas. All milestone dates remain relational ActivityStage records.
- There is no documents table and no audit_logs business table in the updated MVP. Director reviews, committee votes and activity revisions are explicit workflow entities; Pino technical logs remain outside the business database.

7.2 Entity-Relationship Diagram

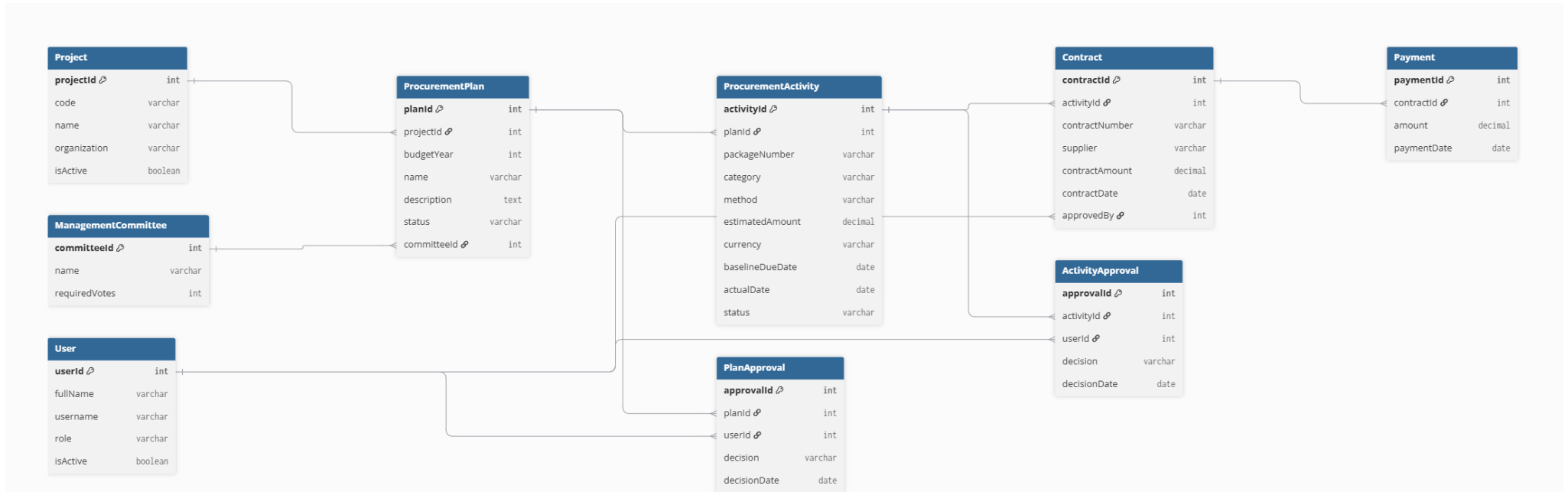


Figure 6. Updated ERD including project assignments and two-level plan approval; document and audit-history entities are removed.

7.3 Core Entity Summary

Entity	Purpose	Key fields / constraints
Organization	Ministry, sector, region, project office or unit hierarchy.	id, parentId, name, type, isActive.
User / Role / UserRole	Authenticated users and one or more explicitly assigned roles.	Unique username/email; only active users sign in.
Project	Director-created project under an organization.	Unique code, name, funding, dates, status.
ProjectAssignment	Authorizes an Officer to work under a project.	Unique project + officer; assignedBy; isActive.
ProcurementPlan	Category-specific plan under one project.	projectId required; category; budgetYear; status; submittedAt; finalApprovedAt.
ProcurementActivity	Main Excel-equivalent procurement record under a plan.	Generated scoped reference; method; amount; categoryData JSONB; status; progress.
ActivityTemplate	Configurable ordered stages for a method and category.	Unique method + sequence; stage name; default days; applicability rules.
ActivityStage	One generated milestone for a Procurement Activity.	Original, revised/current and actual dates; status; delay; remarks.
ActivityRevision	Preserves a replan.	Old/new dates, reason, changedBy, changedAt.
DirectorReview	Stores each Director decision.	Plan, Director, APPROVE/RETURN/REJECT, comment, decision time.
ManagementCommittee / Member	Defines the active five-member approval body.	Exactly five active positions; approval threshold defaults to three.
CommitteeVote	Stores one member decision per plan.	Unique plan + member; APPROVE/REJECT; comment; votedAt.
Supplier / Contract / Payment	Searchable contract and payment-status registry.	Unique contract number; one main contract per activity; paid total cannot exceed current amount.
Alert	User-visible notice for delays, replan events and workflow actions.	Recipient, related record, type, message, status and created time.

7.4 Important Constraints and Indexes

Table	Constraint / index	Reason
project_assignments	Unique(project_id, officer_id); indexes officer_id and is_active.	Fast authorization and My Projects queries.
procurement_plans	Indexes project_id, category, status, created_by.	Project dashboards and approval queues.
procurement_activities	Unique(reference_scope_id, reference_no); indexes plan_id, method_id, status.	Generated reference integrity, search and reports.
activity_stages	Unique(activity_id, sequence_no); indexes status and current target date.	Stable roadmap order and delay queries.
director_reviews	Index plan_id and decided_at.	Decision history and review display.
committee_members	Unique(committee_id, user_id); active-member check.	Exactly one membership per committee and user.
committee_votes	Unique(plan_id, member_id); indexes plan_id and decision.	One vote per member and fast approval count.
contracts	Unique contract_number and unique activity_id.	One main contract per activity and searchable registry.
payments	Indexes contract_id, status and payment_date.	Balance and payment reports.
alerts	Indexes recipient_id, status and created_at.	Role dashboards and unread counts.

7.5 Category-Specific Data Model

The design avoids one table with dozens of mostly empty category columns. Common fields are stored in ProcurementActivity. Extra category information is stored in categoryData and validated against one of four server-side schemas. Stage dates are never stored inside the JSON object; they are normalized into ActivityStage rows so delay and reporting logic remains consistent.

Schema	Examples of validated category data
GoodsActivityData	lotNumber, qualification, domesticPreference, letterOfCreditApplicable, deliveryTerms.
WorksActivityData	lotNumber, qualificationType, domesticPreference, physicalProgress, handoverRequired.
NonConsultingActivityData	serviceType, marketApproach, acceptanceRequired.
ConsultancyActivityData	contractType: Lump Sum/Time Based, firmOrIndividual, selectionMethod details.

8. Validation and Business Logic Design

Rule area	Implementation
Project assignment	Before creating or updating a plan, the service verifies an active ProjectAssignment for the signed-in Officer. The check is repeated on every protected request.
Plan category	A plan has exactly one category. Category cannot change after activities exist without an authorized migration/change process.

Rule area	Implementation
Submission completeness	A plan must contain at least one Procurement Activity; required common/category fields and planned mandatory stages must be valid.
Generated references	ReferenceNumberService builds plan/activity references from configured project/year/category/method/sequence values inside a transaction. The user cannot edit them.
Director review	Only a Director may Approve, Return or Reject a plan in SUBMITTED_TO_DIRECTOR status. Return and Reject require comments.
Committee membership	Only active members of the configured five-member committee may vote.
Committee voting	Database uniqueness prevents a second vote from the same member. Final approval is set when the approval count reaches three.
Unresolved rejection rule	The system does not infer final rejection from Reject votes until the Ministry confirms a rule.
Stage completion	Completed requires actual date. Actual dates must follow configured predecessor rules.
Replanning	Changing a target date requires a reason, creates ActivityRevision, preserves the original date, recalculates delay and creates a Director alert.
Financial rules	Amounts are non-negative; payment amount is positive; total paid cannot exceed current contract amount; original contract amount is preserved.
Controlled values	Only active categories, methods, funding sources, statuses and templates can be selected for new records.

8.1 Derived Calculations

Effective target date = revised date when present; otherwise original planned date.

Delay days = max(0, comparison date - effective target date), where comparison date is the actual completion date for a completed-late stage or the current date for an incomplete overdue stage.

Progress percentage = completed mandatory applicable stages / total mandatory applicable stages x 100. Not Applicable stages are excluded.

Final approval = approval count >= 3. The calculation and status update occur in the same database transaction as the vote.

9. Ethiopian and Gregorian Calendar Design

Concern	Design
Storage	Store one canonical Gregorian date in PostgreSQL.
Entry	The DatePairField accepts either Ethiopian or Gregorian input and displays the converted equivalent before saving.
API	Requests use ISO Gregorian yyyy-MM-dd; responses may include labeled Ethiopian display values.
Validation	Server conversion rejects invalid Gregorian dates and invalid Ethiopian month/day values, including Pagumen rules.
Time zones	Business dates have no time zone. Technical timestamps use UTC and display in the configured Ministry time zone.
Testing	Use approved date pairs covering year boundaries, leap-year Pagumen and representative workbook dates.

10. API Design

10.1 Conventions

- Base path /api/v1; JSON uses camelCase; business dates use ISO Gregorian format.
- List endpoints support paging, sorting and approved filters.
- State-changing requests require authenticated session, same-origin checks and CSRF protection.
- Errors use a standard response with type, title, status, traceId and field errors.

10.2 Main Endpoints

Method and endpoint	Actor	Purpose
POST /auth/login; POST /auth/logout; POST /auth/change-password	All authenticated actors	Session lifecycle.
GET /projects; POST /projects; PUT /projects/{id}	Director	List, create and update projects.
GET /projects/{id}/assignments; POST /projects/{id}/assignments; PATCH /assignments/{id}	Director	Assign or deactivate Officer assignment.
GET /my-projects	Officer	Return only active assigned projects.
POST /plans; GET /plans; GET /plans/{id}; PUT /plans/{id}	Officer; Director view	Create and maintain category-specific plans.

Method and endpoint	Actor	Purpose
POST /plans/{id}/activities	Officer	Create a Procurement Activity, generated reference and stage list.
POST /plans/{id}/submit	Officer	Validate and submit to Director.
POST /plans/{id}/director-review	Director	Approve, Return or Reject.
GET /plans/{id}/approval-summary	Officer, Director, Committee	Show submission, Director decision and committee progress.
POST /plans/{id}/votes	Committee Member	Submit one Approve or Reject vote.
PATCH /stages/{id}	Officer	Update stage status, actual date and remarks.
POST /stages/{id}/replan	Officer	Save revised date and reason; alert Director.
GET/POST /suppliers; GET/POST/PUT /contracts; POST /contracts/{id}/payments	Officer; Director view	Contract and payment registry.
GET /dashboard; GET /alerts	Role scoped	Role-specific summaries and alerts.
GET /reports/{code}/preview; GET /reports/{code}/export	Officer, Director	Preview and export PDF/XLSX.
GET/POST/PATCH /admin/...	Administrator	Users, roles, committee, lookups, methods, templates and references.

10.3 Key Request Models

Model	Important fields
PlanCreateInput	projectId, name, category, budgetYear, Gregorian period, defaultCurrency, description.
ProcurementActivityCreateInput	planId, description, methodId, amount, currency, funding/review fields, categoryData, plannedStages.
DirectorReviewInput	decision: APPROVE RETURN REJECT, comment.
CommitteeVoteInput	decision: APPROVE REJECT, comment.
StageUpdateInput	status, actualDate, remarks.
ReplanInput	newDate, reason, optional impactNote.
ContractInput	activityId, contractNumber, supplierId, original/current/final amounts, dates, status.
PaymentInput	paymentType, amount, paymentDate, status, remarks.

11. User Interface Design

11.1 Navigation and Main Screens

Role	Navigation
Procurement Officer	Dashboard; My Projects; My Plans; Procurement Activities; Contracts; Alerts; Reports.
Procurement Director	Dashboard; Projects; Plans for Review; Procurement Plans; Procurement Activities; Committee Progress; Alerts; Contracts; Reports.
Management Committee Member	Dashboard; Plans for Review; My Decisions.
Administrator	Dashboard; Users; Committee Members; Organizations; Procurement Methods; Activity Templates; Controlled Values; Reference Settings.

11.2 Essential Screen Set

Screen	Essential behavior
Sign In	Username/email, password, clear errors, role-aware redirect. No public registration.
Director Project List/Form	Create project, funding and key dates; assign Officers.
Officer My Projects	Only actively assigned projects; Create Plan inside project context.
Create Plan	Select Goods, Works, Non-Consulting or Consultancy; auto-fill project and Officer; save draft.
Plan Detail	Activities, summary, approval and reports tabs. No documents or audit-history tabs.
Activity Form	Common fields plus category-specific fields; method selection generates planned stages.
Procurement Roadmap	Original, revised/current and actual dates; status; delay; remarks; replan.
Director Review	Read-only plan/activity summary with Approve, Return and Reject.
Committee Review	Read-only plan and totals; Approve/Reject and one comment; voting progress.
Contract and Payment	Linked to an Activity; amounts, dates, payments and balance.
Report Center	Select report, filters, preview, PDF and Excel export.
Administration	Users, roles, committee, methods, templates and reference pattern.

11.3 Category-Specific Activity Forms

Category	Fields shown in addition to common activity fields	Generated milestone emphasis
Goods	Goods/lot description; qualification; domestic preference; optional letter-of-credit applicability; delivery terms.	Bid document, advertisement/invitation, opening, evaluation, award, signing, optional LC, delivery and handover.
Works	Works/lot description; qualification type; domestic preference; handover requirement.	Bid document, opening, evaluation, award, signing, commencement, completion, handover and physical progress.
Non-Consulting Services	Service type; market approach; acceptance requirement.	Request document, invitation, bids/quotations, comparison/evaluation, award, signing, service start/completion and acceptance.
Consultancy Services	Assignment, Lump Sum/Time Based, firm or individual, selection method.	TOR, EOI, shortlist, RFP, technical/financial evaluation, negotiation, draft contract, award, signing and completion.

11.4 Method-to-Template Mapping

Method family	Template
ICB / NCB / RFB	RFB stage template.
Shopping / RFQ	RFQ stage template.
Direct / DC / DIR	Direct Procurement template.
UN	UN Agency template.
QCBS / FBS / LCS	Consultancy Firm template.
CQS	Consultant Qualification template.
INDV	Individual Consultant template.

12. Auto-Fill and Derived Values

Value	Source	Editable
Project, organization, funding and assignment context	Selected assigned project	Only fields explicitly allowed by the plan.
Responsible Officer	Active ProjectAssignment and signed-in Officer	No for normal Officer flow.
Plan category and fiscal year	Parent plan	No on child Activity.
Plan/activity reference	ReferenceNumberService	No.
Category fields	Plan category schema	Officer enters only applicable fields.
Activity stages	Selected procurement method template	Stage names/order not freely edited; optional stages may be Not Applicable.
Project/activity data in contract	Selected Procurement Activity	Reference and responsible Officer read-only.
Progress, delay, paid total and balance	Server calculations	No.
Report title, headings and filters	Selected report and active filter values	No, except permitted report subtitle/remarks.

13. Dashboard and Alert Design

13.1 Officer Dashboard

- Assigned projects; draft, returned, submitted, committee-review and finally-approved plans.
- Activities needing planned dates, delayed stages and progress updates.
- Quick links to continue a plan, update a roadmap, register a contract and generate a report.

13.2 Director Dashboard

- Active projects and assignments; plans awaiting review; returned plans; committee-review progress; finally-approved plans.
- Recent replanning alerts with original date, revised date, reason and Officer.
- Ministry-wide filters by project, category, method, funding source, Officer and status.

13.3 Committee Dashboard

- Plans available for review; own submitted decisions; current approval, rejection and not-voted counts.

13.4 Alert Types

Alert	Recipient / trigger
PLAN_SUBMITTED	Director when an Officer submits a plan.
PLAN_RETURNED / PLAN_REJECTED / DIRECTOR_APPROVED	Officer after Director decision.
COMMITTEE_REVIEW_READY	Five active committee members after Director approval.
PLAN_FINALLY_APPROVED	Officer and Director when approval count reaches three.
ACTIVITY_REPLANNED	Director whenever an Officer replans a stage.
OVERDUE_STAGE / CONTRACT_DUE	Scoped Officer and Director based on configured thresholds.

14. Report Design

Reports are generated from the same relational data and calculation services used by dashboards. The user does not re-enter information for reporting. Report column mappings reproduce the relevant MoA Excel formats while using consistent system fields.

Report	Main content
Annual Procurement Plan	Category-specific activity reference, description, amount, funding, method and planned milestones.
Plan vs Actual	Original, revised/current and actual dates by stage, status, delay and remarks.
Procurement Step Report	Method-specific RFB, RFQ, Direct, UN, QCBS/FBS/LCS, CQS or INDV columns.
Delayed Procurement	Activity, delayed stage, Officer, effective target, delay days, status and remarks.
Monthly / Quarterly Summary	Counts and values by category, method and Treasury/Loan/Grant with subtotals and totals.
Contract and Payment Status	Supplier, contract amounts, variations, total paid, payment components, balance, dates and status.
Project / Officer Summary	Counts, amounts, progress and delays by project or assigned Officer.
Approval Status	Plan submission, Director decision, five member votes and final approval status.

14.1 Export Rules

- PDF and Excel include report title, selected filters, generation date and totals.
- Amounts always show currency; mixed currencies are not added into one unlabeled total.
- Dates may be exported in Ethiopian, Gregorian or both formats.
- Not Applicable is displayed for stages that do not belong to the selected method.
- Wide plan-vs-actual and step reports use landscape PDF and preserve row-level Excel data.

15. Security and Authorization Design

15.1 Authentication

- Argon2id password hashes; secure HTTP-only session cookie; HTTPS; configurable inactivity expiry.
- Administrator-created users must change temporary password at first sign-in.
- Configurable lockout for repeated failed attempts; CSRF protection on state-changing requests.

15.2 Permission Matrix

Permission	Officer	Director	Committee	Admin
Create/update project	No	Yes	No	No
Assign Officer to project	No	Yes	No	No
Create plan/activity in assigned project	Yes	View	View approved	No
Submit plan	Yes	No	No	No
Director review	No	Yes	No	No
Committee vote	Only if separately assigned	Only if separately assigned	Yes	No
Update stages and replan	Yes	View	View	No
Register contracts/payments	Yes	View	View plan summary	No
View dashboard/reports	Scoped	Ministry-wide	Committee scope	Administration status
Manage users/configuration	No	No	No	Yes

15.3 Scope Enforcement

Hiding UI actions is not sufficient. Express authorization middleware and application services repeat project-assignment, role, plan-status and record-scope checks. Prisma queries include the permitted project or organization scope. Direct object access outside scope returns 403 without exposing record details.

16. Error Handling and Technical Logging

Case	User response
Validation failure	400 with field-specific messages; entered data remains in the form.
Unauthenticated	401 and redirect to Sign In.
Unauthorized or unassigned project	403 with a clear permission message.
Not found	404 with safe record-not-found message.
Duplicate reference/contract	409 conflict; server does not expose database internals.
Invalid plan state	409 explaining the required workflow status.
Duplicate committee vote	409: the member has already submitted a decision.
Payment exceeds balance	409 with remaining amount.
Export failure	Safe message with correlation ID and retry instruction.

16.1 Technical Logs

Pino writes structured technical logs containing timestamp, severity, correlation ID, route, response status and safe diagnostic context. Logs must not contain passwords, session secrets or unnecessary procurement personal data. Technical logs are accessible to authorized ICT staff only and are not exposed as a business audit-history screen.

17. Deployment Design

Component	Deployment
Nginx	HTTPS termination, secure headers, request limits and routing.
Next.js frontend	Standalone container serving the role-aware UI.
Express backend	Container running REST services, Passport sessions and scheduled jobs.
PostgreSQL	Persistent database service with restricted network access.
Report temporary path	Ephemeral or managed temporary directory cleared after export/download.
Logs	Rotated files or MoA-approved central log destination.

17.1 Required Configuration

NODE_ENV, DATABASE_URL, SESSION_SECRET, FRONTEND_URL, API_PORT, CORS_ALLOWED_ORIGIN, SESSION_MAX_AGE_MINUTES, DISPLAY_TIME_ZONE, ALERT_UPCOMING_DAYS, REPORT_TEMP_PATH, LOG_LEVEL and BACKUP_TARGET_PATH.

17.2 Deployment and Rollback

1. Build and test tagged frontend and backend images.
2. Create a database backup before deployment.
3. Apply reviewed Prisma migrations.
4. Deploy through Docker Compose and run health, login, role, assignment and database checks.
5. Smoke-test project assignment, plan submission, Director review, committee vote, stage update, contract and report export.
6. Rollback containers and restore the pre-deployment database if a migration is not backward compatible.

18. Backup and Recovery Design

Item	Design
Frequency	Daily PostgreSQL backup and mandatory pre-deployment backup.
Method	pg_dump custom format plus protected configuration backup.
Location	MoA ICT-protected storage separate from the application server.
Retention	Initial 7 daily, 4 weekly and 6 monthly backups, subject to MoA policy.
Verification	Daily success check and scheduled restore test to non-production.
Recovery target	Initial RPO 24 hours and RTO 4 hours after infrastructure is available.
Scope change	No permanent procurement-document directory is included in backup because upload/storage is outside the updated MVP.

19. Testing Approach

Test type	Updated coverage
Unit	Reference generation, assignment authorization, plan state rules, approval count, one-vote rule, delay, progress, payment balance and calendar conversion.
API integration	Authentication, project creation/assignment, plan submission, Director decisions, committee votes, activity/replan, contracts/payments, filters and safe errors.
Database	Migrations, scoped uniqueness, foreign keys, one vote/member, project assignment uniqueness, payment checks and transaction rollback.
Frontend	Role navigation, category-specific forms, method-generated stages, validation, read-only review screens and approval progress.
Role access	Officer, Director, Committee, Administrator and unauthenticated matrix for every protected screen and endpoint.
Report	Approved category/method columns, filters, totals and PDF/XLSX readability reconciled to sample Excel results.
Calendar	Approved Ethiopian/Gregorian pairs including year and Pagumen boundaries.
Performance	Normal page/search/save operations and standard reports against agreed UAT thresholds.
UAT	End-to-end flows from project assignment through final approval, progress tracking, contract status and reporting.

19.1 Minimum End-to-End Test Scenarios

- Director creates a project and assigns an Officer; an unassigned Officer is denied plan creation.
- Officer creates each of the four plan categories and sees only the applicable activity fields.
- Method selection generates the correct ordered stages and a unique server-generated reference.
- Officer submits a complete plan; Director returns it; Officer corrects and resubmits.
- Director approves; exactly five configured committee members receive access.
- Two approvals keep the plan in Committee Review; the third approval changes it to Finally Approved in the same transaction.
- A duplicate committee vote is rejected.
- Officer replans an approved stage with a reason; the original date remains and the Director receives an alert.
- Contract and payment totals reconcile and payment above current contract amount is rejected.
- PDF and Excel outputs match the selected report preview and source data.

20. Traceability, Decisions and Completion Conditions

20.1 Updated Requirement-to-Design Traceability

Updated requirement	Design implementation
Director creates projects and assigns Officers	Project/Assignment module, project_assignments table, Director UI and endpoints.
Officer plans only in assigned project	AuthorizationService assignment check, /my-projects, plan create validator and DB relationship.
Four plan categories with different fields	Plan.category, category-specific Zod schemas, conditional UI forms and report mappings.
Method-specific activity stages	procurement_methods, activity_templates and stage generation transaction.
Officer submits to Director	Plan submission service and SUBMITTED_TO_DIRECTOR state.
Director approves/returns/rejects	DirectorReview entity, endpoints, UI and state-transition service.
Five-member committee and 3-of-5 rule	ManagementCommittee, CommitteeMember, CommitteeVote and ApprovalStatusService.
Generated unique references	ReferenceNumberService and scoped unique database constraint.
Director alert on replan	ActivityRevision and AlertService transaction.
No documents or audit history	No document-storage module/table/API/UI and no business audit-history module/table/API/UI.
Excel-based planning and reporting	Category/method schemas, roadmap, report mappings, PDFKit and ExcelJS exporters.

20.2 Open Decisions Requiring MoA Confirmation

Decision	Reason
Committee rejection rule	The current clarification defines final approval at three approvals but does not define automatic final rejection, abstention or voting deadline.
Committee scope	Confirm whether one committee applies Ministry-wide or membership varies by project/plan.
Multiple roles	Confirm whether a Director or Officer may also act as a committee member using the same account.
Approved-plan amendments	Confirm whether major changes after final approval restart the full Director and committee approval cycle.

Decision	Reason
Reference pattern	Confirm official plan/activity pattern and organizational uniqueness scope.
Official report columns	Procurement Directorate must approve final column labels/order for each Goods, Works, Non-Consulting and Consultancy report.
Terminology	Confirm whether the user-facing main record should be "Procurement Activity" only or "Procurement Activity / Package".

20.3 Design Completion Conditions

- MoA approves the four-actor workflow and committee rules.
- Procurement Directorate confirms category schemas, method-stage templates and official report columns.
- ICT approves the selected stack, deployment, session, backup and technical-log configuration.
- Prisma migrations, OpenAPI specification, UI wireframes and test cases are generated from this approved design.
- Any later change that alters the approved workflow is recorded in the requirements baseline before implementation.

Document boundary: This SDD defines implementation design. Complete source code, migration files, generated OpenAPI output, detailed test cases, deployment commands and user manuals remain separate development artifacts.